TEAM TOUCANET · KASHMIR WORLD FOUNDATION

Playback Experiment: Progress Review

Is the response to simulated human presence a biological change in animal calls, or only the playback sound?

Rashi Tyagi · November 5, 2026

Repository: **github.com/ratyagi/toucanet-playback-sensitivity**

Status: draft. The hypotheses file is a reconstruction. The listening review is pending.

CONTEXT

The question and the method

- **Question:** After a simulated human passes a recorder, do animals change their calls? Or does the detector only hear the playback?
- **Data:** 6 AudioMoth recorders, 5 days, 29 simulated human-presence events (11 isolated, 11 overlapping, 7 incomplete).
- **Measure:** Call rate after each event, minus a baseline for the same recorder and time of day. Bins are 15 minutes.
- **Test:** Random-intercept model by recorder. The gate is two-sided p below 0.05 for the original 0-60 minute analysis.
- **Controls:** A placebo-in-time test (1,000 draws) and variants that remove clips that YAMNet flagged as human sound.
- **Rules fixed first:** `PREREGISTRATION.md` was committed before any variant ran. Every later change is in `DEVIATIONS.md`.

COMPLETED AND TESTED

Three parts of the work are done

1 Reproducible setup, checked against the lead's result

- Pinned Python 3.12 environment. The dataset is checked with SHA-256.
- The lead's result reproduced exactly on two machines: estimate -0.41, standard error 0.87.
- 29 events rebuilt from clip times: onset = clip time minus `Sim Relative Time`.

2 Six variants, a placebo test, and the decision rules

- Six windows in JSON configs. Variants 4a and 4b remove YAMNet-positive clips and the 300 seconds after each.
- Placebo-in-time: 1,000 draws, seed 20251002, strict and relaxed candidate pools, position-matched clip drops.
- Gate and decision rules coded in `src/decision.py` and tested.

3 Diagnostics, review sample, and reporting

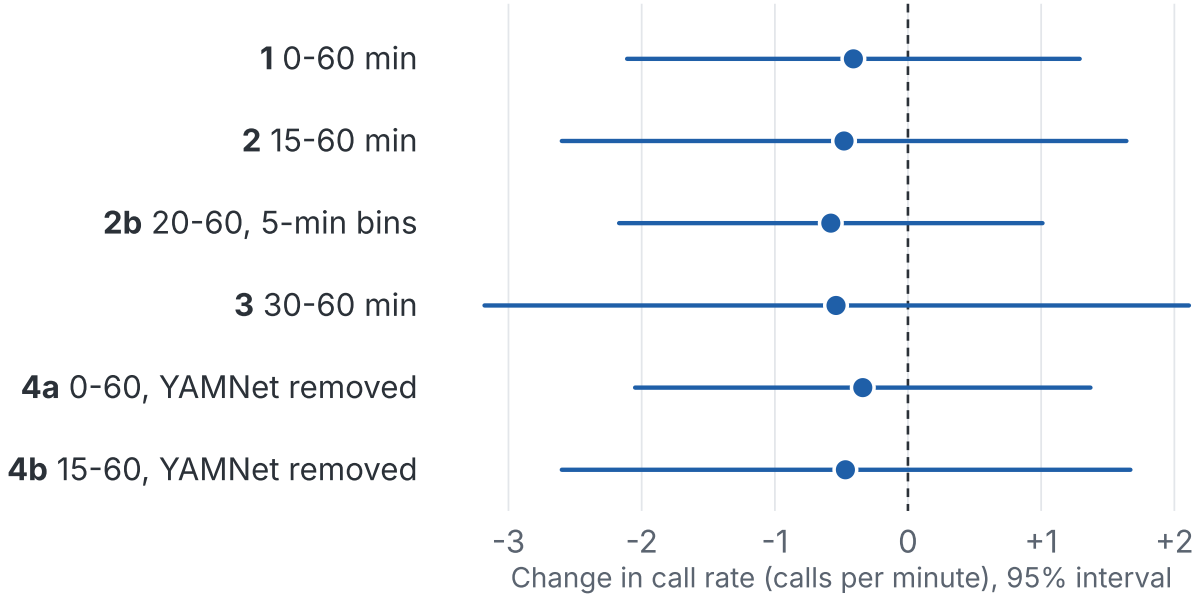
- Six per-recorder figures and the 22-versus-7 YAMNet split. A blind listening sample of 200 clips, with weights.
- Unit tests for every module. The report is built from the result tables by script.

RESULTS

Result 1: the pre-registered test is inconclusive

Gate fails: p = 0.64. It needs p < 0.05.

- All six variants give a small drop of 0.34 to 0.58 calls per minute.
- Every interval includes zero.
- With 10 events, the test cannot detect changes smaller than 1.6 to 2.6 calls per minute.
- This is not evidence of no effect.



RESULTS

Results 2 and 3: exploratory checks

2 The sign depends on the baseline EXPLORATORY

- Without baseline adjustment the estimate is +0.68 (p = 0.28, 11 events). With complete windows only it is +0.53 (5 events).
- Audio_Moth_1 to 4 average +0.66. Audio_Moth_5 and 6 average -1.91. Each group has 5 events.
- Audio_Moth_5 and 6 have no presence-free days, so their baselines are thin.

3 The placebo test shows no drop EXPLORATORY

- 5 events on 4 recorders. Observed +0.47. The placebo 95% range is -1.36 to +0.47. Two-sided p = 0.054.
- The one-sided test for a drop gives p = 0.97. The six variants share the same 5 events.
- YAMNet flagged only 1.6% of simulated clips (168 of 10,636). Mean contrast: -0.36 with a detection, -0.32 without. The split is confounded with recorder.

REVIEW AND APPROVAL

Three decisions I would like you to review

1 The count model and two extra checks

- Count model: negative binomial, recorded-minutes offset, recorder fixed effects, standard errors clustered by recorder and date. Six recorders are too few for a random effect.
- Leave-own-day-out baseline, and a leave-one-recorder-out refit. All three are exploratory and cannot change the verdict.
- The specifications were committed before any run.

2 The design of the listening review

- I review 200 clips (100 YAMNet-positive, 100 negative), blind to the label and the position. Order is random.
- Clips that overlap another simulated sound are excluded (2,400 clips). Only 104 positive clips remain, so the positive sample is almost all of them.

3 Three inputs from you

- Your `hypotheses.json`. I used a reconstructed draft and labeled every result as draft.
- Permission to keep `event_study.py` and `placebo_hypothesis_analysis.py` in the repository.
- Whether `Sim Relative Time` 0 is the start or the middle of the playback.

ISSUES AND NEXT STEPS

Three issues and my plan for each

1

The baselines are thin

- **Issue:** Audio_Moth_5 and 6 hold 16 of 29 events and have no presence-free days. The baseline also includes the event's own day.
- **Next:** Run the leave-own-day-out check. Ask for recordings from days without playback at those sites.

2

The model fits are fragile

- **Issue:** 31 of 72 fits used the fallback optimizer, and one had no valid standard error. The plan lists a count model that is not fitted yet.
- **Next:** Fit the count model and the leave-one-recorder-out refit. Report them even if they disagree.

3

Two steps need other inputs

- **Issue:** The listening review needs the audio folder. The final report needs your hypotheses file.
- **Next:** Finish the review. Add a go, no-go, or hold recommendation to the report. Hand the verdict to the team for the December LODO comparison.

REPOSITORY

Everything is in GitHub and reproducible

- **Link:** github.com/ratyagi/toucanet-playback-sensitivity
- **Run it:** Steps 1 to 11 in `README.md` rebuild every table, figure, and the report. All seeds are fixed.
- **Plan and rules:** `PREREGISTRATION.md` (fixed first) and `DEVIATIONS.md` (every later change).
- **Results:** `REPORT.md` and `results/tables/`. Figures are in `results/figures/`.
- **Code and tests:** `src/` and `tests/`. The lead's two scripts are in `src/lead/`, unchanged.
- **Data:** The raw dataset is not in git because it is large. Its SHA-256 is in `reference/dataset_checksum.txt`.

29
events rebuilt and checked

6
pre-registered variants

1,000
placebo draws, fixed seed